

Lab 8: Fast Decay with RC Circuits

March 27, 2026

Jaysen Tsao

Lab Partner: Alex Yue

PHYS 262 CRN 30949

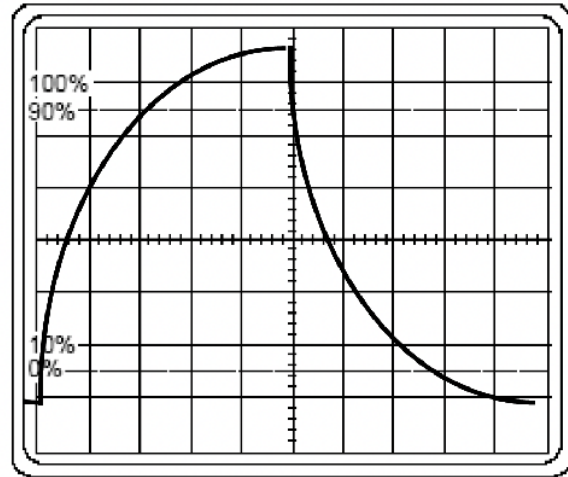
Professor Szwerc

Pre-lab Questions

1. You have constructed an RC circuit and you see the trace to the right on your oscilloscope screen. You have set the vertical scale to 1 V per square and the horizontal to 20 ms per square. What is the time constant of the RC circuit?

The time constant $\tau = RC$ is equivalent to the “1/e life” of the voltage across the capacitor. That is, at what time t after discharging does the voltage across the capacitor drop to $1/e \approx 36.8\%$ of its initial value?

The range of the voltage is from the line at -10% to the line at 110% . $1/e \approx 36.8\%$ between -10% and 110% is approximately 34.2% , which intersects with the curve about 1 division past the y-axis.



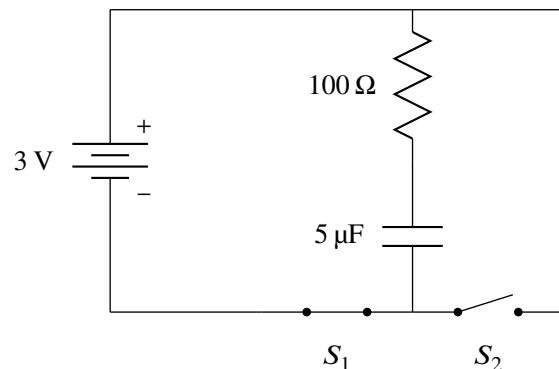
This corresponds to a time constant of approximately:

$$\tau \approx 1 \text{ division} \cdot 20 \text{ ms/division} = 20 \text{ ms.}$$

2. The RC circuit is constructed as shown on the right.
 - b. You then change the resistance to 200Ω . How does the time constant change?

The time constant is proportional to the resistance: $\tau \propto R$. Thus, doubling the resistance will double the time constant:

$$\tau' = 2\tau.$$



- b. You then change the capacitance to $20 \mu\text{F}$. How does the time constant change?

The time constant is proportional to the capacitance: $\tau \propto C$. Thus, quadrupling the capacitance will quadruple the time constant:

$$\tau' = 4\tau.$$

- c. You then change to a 9 V battery. How does the time constant change?

The time constant is independent of voltage, so the time constant will not change:

$$\tau' = \tau.$$